

Krishna Sudhir Malgi

Software Engineer — Python Full Stack Developer — FastAPI — REST APIs — SQL

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Professional Summary

Software Engineer with hands-on experience building full-stack web applications using **Python**, **FastAPI**, **JavaScript**, and **SQL**. Skilled in developing **REST APIs**, designing relational databases, and building **Machine Learning** and **Retrieval-Augmented Generation (RAG)** applications with **OCR** and **LLM** integration. Strong foundation in **OOP**, **Operating Systems**, **Computer Networks**, **PostgreSQL**, and **MySQL**, with exposure to deployment using **Render** and **Vercel**.

Education

Jawaharlal Nehru National College of Engineering (JNNCE), Shimoga

Dec 2022 – Jun 2026

B.E in Information Science and Engineering

CGPA: 8.08 / 10

Projects

ML-Based Food Recommendation System for Health Conditions– [GitHub]

Aug 2025 – Nov 2025

Python, Machine Learning, KNN, FastAPI, PostgreSQL, Streamlit

- Designed a **KNN** recommendation algorithm on **1,000+** food entries, filtering to **956** valid items across **9 medical conditions** (Diabetes, Hypertension, Heart, Kidney, PCOD, etc.) using **13 nutritional attributes**.
- Engineered a hybrid scoring system (**60% cosine similarity, 40% percentage-based match**) with **StandardScaler** normalization to prevent high-magnitude features like Sodium from biasing results.
- Built and deployed **FastAPI** REST APIs with a **Streamlit** frontend; designed a normalized **PostgreSQL** schema for real-time nutritional filtering and ranking.

Multi-Modal RAG Chatbot for Document Q&A– [GitHub]

Jan 2026 – Mar 2026

Python, FastAPI, pgvector, Supabase, Groq LLM, OCR

- Architected a full-stack **RAG** platform processing PDFs, scanned documents, and images using **Tesseract OCR** and **Camelot** table extraction, with token-by-token streaming via **Server-Sent Events (SSE)**.
- Designed a hybrid semantic search pipeline using **pgvector** and **all-MiniLM-L6-v2** embeddings with query expansion and re-ranking, serving responses via **Groq's LLaMA-3.3-70B**.
- Implemented **Supabase JWT authentication** with per-request validation; enforced **zero-data-retention** by purging user vector embeddings on logout.

Publications

- C. Sagar, K. Bambalwadi, K. Sathish, **Krishna Sudhir Malgi**, Dr. R. Sanjeev Kunte, “*Machine Learning Approach for Smart Food Recommendation System*,” **JETIR**, ISSN: 2349-5162.

Internship Experience

Full Stack Python Development Intern, Kodnest Technologies

Feb 2026 – Present

- Build and maintain full-stack web applications using **Python**, **HTML**, **CSS**, and **JavaScript**; design relational **MySQL** schemas and develop **REST APIs**.
- Collaborate using **Git/GitHub** version control; use **GenAI-assisted development** to speed up debugging and learning of new frameworks..

Data Analytics Virtual Intern, Deloitte – Forage

Jul 2025

- Performed **data cleaning, analysis, and visualization** on business datasets to support data-driven decisions.

Technical Skills

Languages: Python, JavaScript, SQL

Frontend: HTML, CSS

Backend: Python, FastAPI, REST API Development

Databases: PostgreSQL, MySQL, pgvector, Supabase

AI/ML: Machine Learning, KNN, RAG Pipelines, OCR, LLM Integration, Data Preprocessing

Tools & Platforms: Git, GitHub, Render, Vercel, VS Code

Core CS: DSA, OOP, Operating Systems, Computer Networks

Certifications

- Python Programming: Zero to Hero** – Udemy
- FastAPI Course** – CampusX
- Introduction to Generative AI**

Languages

Languages: English, Hindi, Kannada, Konkani